

Exploratory Data Analysis (EDA) & Data Cleaning Report

1. Dataset Overview

Detail	Raw Dataset (12 PPana.csv)	Cleaned Dataset (df_clean.xlsx)
Rows	50,000	49,152
Columns	7	7
Missing Values	18,713	0
Duplicates	548	0

2. Column Comparison

Column	Raw Data Type	Cleaned Data Type	Notes
BillNo	int64	int64	Unchanged
Itemname	object	object	Unchanged
Quantity	int64	int64	Outliers handled
Present_Date	object	datetime64	Converted to datetime
Price	float64	float64	Cleaned and standardized
CustomerID	float64	int64	Converted to integer
Country	object	object	Unchanged

3. Summary Statistics Comparison

Metric	Quantity (Raw)	Quantity (Cleaned)	Price (Raw)	Price (Cleaned)
Mean	8.60	8.85	4.55	4.60
Std Dev	37.73	35.06	63.76	64.31
Min	-2600	1	0	0.07
Max	2880	2880	13541.33	13541.33

4. Data Cleaning Summary

Cleaning Step	Description
Handling Missing Values	Removed rows with missing data in critical columns.
Duplicate Removal	Removed 548 duplicate transactions.
Data Type Conversion	Converted date and ID columns to correct types.
Outlier Treatment	Removed negative or invalid quantities.
Standardization	Ensured consistent naming and format across columns.

5. Key Insights (Post-Cleaning)

Top Product: **WHITE HANGING HEART T-LIGHT HOLDER** remains the best-seller.

Sales Quantity: Majority transactions involve 1–8 units.

Time Range: Data spans from **Dec 2010** to **Jan 2011**.

Country: Primary market observed in the UK.